

LESSON 7-7 • SESSION 1 • GUIDED NOTES

Polygons and Perimeter on the Plane

Learning target – I can find the perimeter and area of a polygon drawn on the coordinate plane. · 6.G.A.3

WORDS TO KNOW*fill in the missing words*

WORD	WHAT IT MEANS	EXAMPLE
polygon <i>polígono</i>	A closed shape with _____ sides.	triangle, rectangle, square
vertex <i>vértice</i>	A _____ point of a polygon.	(-2, 3)
perimeter <i>perímetro</i>	The distance all the way _____ a shape.	add every side length
area <i>área</i>	The number of _____ units inside a shape.	rectangle: length × width

KEY IDEA*complete each sentence*

- Two vertices that share a coordinate make a _____ or vertical side.
- Perimeter is the ____ of all the sides. Area of a rectangle is length ____ width.

WORKED EXAMPLE*fill in the blanks as we solve*A rectangle has vertices $(-1, 4)$, $(5, 4)$, $(5, -3)$ and $(-1, -3)$. Find its perimeter and area.**Step 1** Top side: same y, opposite signs, so $1 + 5 = \underline{\hspace{1cm}}$ units.**Step 2** Right side: same x, opposite signs, so $4 + 3 = \underline{\hspace{1cm}}$ units.**Step 3** Perimeter: $6 + 7 + 6 + 7 = \underline{\hspace{1cm}}$ units.**Answer** Area = $6 \times 7 = \underline{\hspace{1cm}}$ square units**YOU TRY***use the example as a model*

- A square has vertices $(-3, 2)$, $(1, 2)$, $(1, -2)$ and $(-3, -2)$. Find its perimeter.

*Show your work.***Answer** _____ units

- Three corners of a rectangle are $(-4, 1)$, $(3, 1)$ and $(3, -5)$. Name the fourth corner.

*Show your work.***Corner** _____**SUM IT UP***in your own words*

To find the perimeter of a polygon on the coordinate plane, I ____ because ____.

LESSON 7-7 • SESSION 2 • GUIDED NOTES

Designing Polygons with Scale

Learning target – I can use a scale to find a missing dimension of a figure on the coordinate plane. • 6.G.A.3

WORDS TO KNOW*fill in the missing words*

WORD	WHAT IT MEANS	EXAMPLE
scale <i>escala</i>	How much real length one grid ____ stands for.	1 unit = 4 meters
area <i>área</i>	The flat space ____ a shape, in square units.	20 m × 12 m = 240 m ²
dimensions <i>dimensiones</i>	The ____ and width of a shape.	24 ft by 16 ft

KEY IDEA*complete each sentence*

- Multiply grid units by the ____ to get real length.
- To find a missing side, ____ the area by the known side.

WORKED EXAMPLE*fill in the blanks as we solve*

A rectangle is 5 units by 3 units on a grid. Each unit is 4 meters. Find its real area.

Step 1 Real width: $5 \times 4 = \underline{\hspace{1cm}}$ m.

Step 2 Real height: $3 \times 4 = \underline{\hspace{1cm}}$ m.

Answer Area = $20 \times 12 = \underline{\hspace{1cm}}$ m²

YOU TRY*use the example as a model*

- On a plan, 1 unit = 5 ft.
 - 8 units = ____ ft
 - 35 ft = ____ units
- A rectangle has an area of 600 ft². One side is 20 ft. Find the other side.

Show your work.

Answer _____ ft

SUM IT UP*in your own words*

To find a missing dimension with a scale, I ____ because ____.
